



Research paper

A novel mathematical modeling and simulation of multi-tugboats assisted pushing operations for a barge

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ABSTRACT

This paper proposes a novel mathematical model for the simulation of multi-tugboats assisted pushing operations for a barge, addressing the common simplification in existing studies where tugboats are treated as ideal thrusters, neglecting the influence of the hydrodynamic and environmental disturbances of the tugboats on the pushing force. By assuming fixed relative positions between the tugboats and the barge, the model incorporates the hydrodynamic characteristics and environmental disturbances of the tugboats, using the actual thrust of main thrusters as the control input instead of an idealized pushing force. The proposed low-frequency model integrate mass, damping, and disturbance terms derived from each tugboat, enhancing physical realism without significantly increasing computational complexity. Simulation results with various environmental disturbances demonstrate that, while barge positioning performance remains comparable between the conventional and proposed models, the required main thruster thrust substantially differs from the idealized pushing force due to tugboat self-disturbance compensation. The study further reveals frequent thruster saturation phenomena, indicating that the conventional model should apply a conservative loss rate to pushing force change rates to ensure feasibility. The proposed approach provides a more accurate foundation for controller design and performance evaluation in pushing operations.

1. Introduction

The movement and positioning of barges at sea, as well as the maneuvering of large vessels in port, typically rely on the assistance of tugboats. Depending on the connection methods between tugboats and the barge, tug operations are primarily categorized into two modes: pushing and towing, and the scenarios are shown in Fig. 1. With advances in unmanned vessels and intelligent control technologies, corresponding researches have progressively shifted from single-vessel control to multi-vessels cooperation, prompting extensive exploration by scholars worldwide into intelligent and cooperative control of multi-tugboats systems in recent years.

For towing operations, the tugboats control the movement of the barge via towlines. Hajieghrary et al. (2018) used differential geometry methods to develop unified dynamic models and constraints for multi-vessel cooperative towing systems. Ianagui and Tannuri (2019) proposed a consensus protocol-based cooperative control structure for multiple dynamic positioning vessels. Yun and Jian (2018) addressed the heading control problem of a two-tugboats system towing a barge by developing a coordinated strategy that achieves intelligent and stable steering of the barge through the synchronization of heading angles and towing forces. Du et al. (2021) developed a multi-agent hierarchical control architecture, enabling autonomous tugboats to cooperatively control the motions of the barge with environmental disturbances. Jiang et al. (2023) introduced a fuzzy adaptive approximation-based cooperative consensus control strategy, achieving high-precision towing of a barge by multiple dynamic positioning tugboats under unknown model parameters and towline tension disturbances. Jiang et al. (2025) further proposed two cooperative control methods, strategies—formation-based and force-allocation-based, for unmanned multi-tugboats during towing operations. Ianagui et al. (2025) devised a cooperative control protocol leveraging sliding mode control and super-twisting algorithms for autonomous coordination of multiple dynamic positioning vessels to maintain formations and track trajectories with environmental disturbances.

For pushing operations, tugboats deliver thrust through direct contact against the side of a barge, and thereby control its motions. Feemster and Esposito (2011) introduced an integrated framework combining trajectory generation method, adaptive tracking control, and thrust allocation

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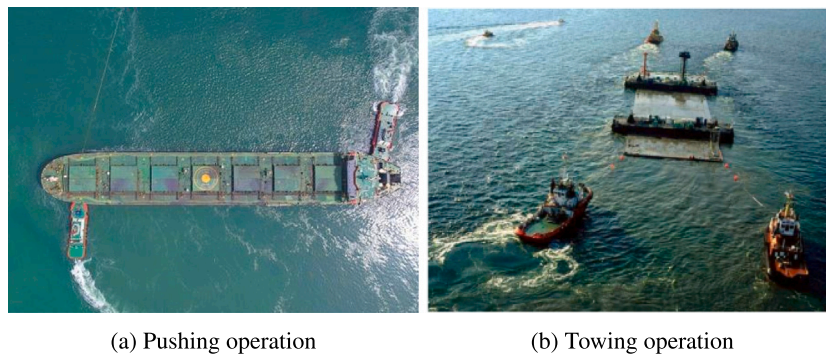


Fig. 1. Pushing and towing operation scenarios.

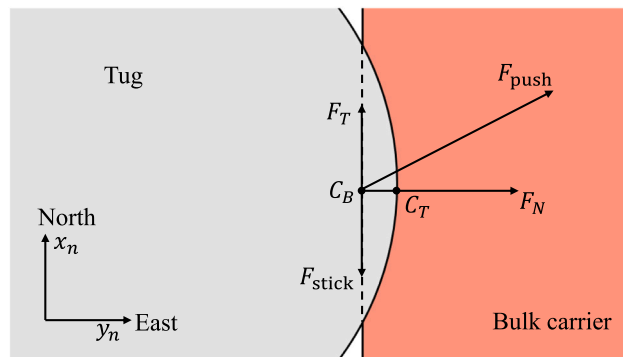


Fig. 2. Tugboat-barge interaction model by Kristiansen et al. (2024).

for autonomous surface vessels. Bui et al. (2011) designed a quadruple-tugboats berthing system based on adaptive control and pseudo inverse thrust redistribution. Van Bui and Kim (2011) developed an autonomous berthing method for tugboats using sliding mode control and constrained thrust allocation to ensure robust trajectory tracking under environmental disturbances. Bidikli et al. (2015) addressed uncertain asymmetric input gain matrices using matrix decomposition, designing robust controllers for tracking of the barge. Sartoretti et al. (2016) proposed a distributed control strategy enabling a fleet of autonomous surface vessels to achieve the manipulation of the barge through conditional enforced caging. Bidikli et al. (2016) further developed robust controllers addressing asymmetric inertia matrices and positional uncertainties for dynamic positioning of a barge. (Chen et al., 2025) proposed a multi-tugboats collaborative pushing operation control strategy based on model predictive control and event-triggered mechanism.

By comparing towing and pushing operations, it can be found that these two methods have their own strengths and limitations. Towing operation poses lower risk and accommodates a wider range of vessel types but offers poorer maneuverability and higher control complexity for the barge. Pushing operation provides superior maneuverability. Consequently, hybrid towing–pushing systems have emerged as a research focus in recent years. Huynh and Young-Bok (2024) proposed a hybrid control system combining towing and pushing tugboats, with a high-order sliding mode observer estimating environmental disturbances for automated floating platform transport. Park et al. (2025) established a general mathematical model for tugboats-assisted operations, utilizing a hybrid H_2/H_∞ control framework to enhance system robustness and tracking performance.

However, in the numerical simulations of pushing operations, all the above studies (Feemster and Esposito, 2011; Bui et al., 2011; Van Bui and Kim, 2011; Bidikli et al., 2015; Sartoretti et al., 2016; Bidikli et al., 2016; Huynh and Young-Bok, 2024; Chen et al., 2025; Park et al., 2025) related to pushing operation universally simplify the dynamic model of tugboats, treating tugboats as thrusters on the side of barge and controlling the barge by pushing force. This marks a key modeling distinction from real towing operations: the towing operation delineates a clear interactive relationship between tugboats and the barge via the towlines and by adjusting the position of tugboats, the towing force exerted on the barge is effectively controlled. By contrast, pushing operations typically treat the tugboat simply as a thruster, neglecting its own dynamics. Because of the environmental disturbances and the hydrodynamic influences of the tugboats, the idea pushing thrust cannot be directly and precisely controlled by the hardware of the tugboats. The overall pushing cooperative control shall treat the actual thrusters of the tugboats as the control output.

Some studies have addressed the calculation of tugboats–barge interaction forces in pushing operations. Kristiansen et al. (2024) calculated the interaction forces between tugboats and the barge by modeling collisions of rubber fenders mounted on the side of the barge, thereby providing essential dynamic models for their in-house simulator. The interaction model is shown in Fig. 2. Similar methodologies are widely applied to investigate collisions between dynamically positioned vessels and jackets during float-over installations. However, this method entails high model complexity and requires shorter simulation time-steps to ensure the convergence, leading past studies to simplify tugboats as thrusters. Kristiansen et al. (2024) also did not further address the motion control problem of tugboats–barge systems.

Sano (2023) proposed a different way to deal with the calculation of pushing force in their research on the simulation of various operation modes such as pushing, towing, and free sailing. In their study, the interaction force between the tugboat and the barge is solved by assuming that the tugboat and the barge are connected by a pin connection, although this assumption is not feasible. The neglected environmental disturbances in their study will give a large tangential force to the bow of the tugboat in reality, resulting in damage to the connection structure. However, their approach to pushing force is still worth reference, because the dynamic positioning system has the possibility to implement this method.

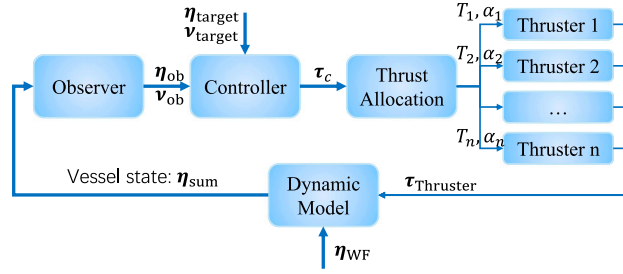


Fig. 3. The simulation procedure of the dynamic positioning vessel.

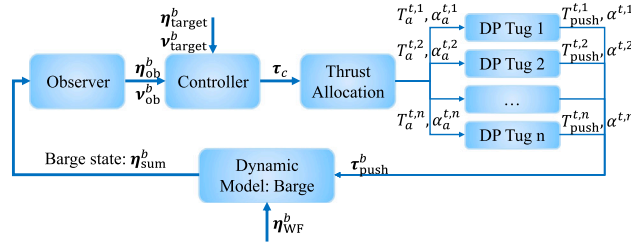


Fig. 4. The simulation procedure of the pushing operation.

Thus, this paper addresses the core issues observed in numerical simulations of pushing operations: the excessive simplification of conventional mathematical models, and the prohibitive complexity of elastic collision calculations. Based on the assumption that the tugboats maintain position relative to the barge, a novel mathematical model that incorporates the influence of environmental disturbances and tugboats hydrodynamics is proposed, using the actual thrust of thrusters as the model inputs. This approach enhances the realism and rationality of the mathematical model without significantly increasing computational cost. The paper details the derivation of proposed mathematical model, presents corresponding simulation results with environmental disturbances, and quantifies the differences. Furthermore, this paper examined how the phenomenon of thruster saturation, observed during the computation of the proposed model, influences the configuration of pushing force parameters in the conventional model calculations.

2. Mathematical modeling of multi-tugboats assisted pushing operation

2.1. Conventional modeling of the dynamic positioning system and pushing operation

The modeling of the dynamic positioning vessel generally follows the procedure illustrated in Fig. 3. In this procedure, the thrust allocation module, upon receiving the target control force from the controller, converts it into thrust and angle commands for each thruster, which are then transmitted to the respective thrusters.

The modeling of the pushing operation generally follows the procedure illustrated in Fig. 4. In existing studies, the calculated results for thrust allocation, $T_a^{t,i}$ and $\alpha_a^{t,i}$, are typically adopted as the pushing force and orientation generated by the tugboat, denoted as $T_{push}^{t,i}$ and $\alpha^{t,i}$. In other words, the tugboats are modeled as side thrusters for the barge within the context of dynamic positioning numerical modeling. By comparing Figs. 3 and 4, it can be observed that there is no fundamental difference, aside from the control input designation within the dynamic model changing from thruster force to pushing force. However, in a tugboats–barge system, the actual actuators are the thrusters of tugboats. Considering environmental disturbances and hydrodynamic issues, the thrust generated by the thrusters does not exactly equal the pushing force exerted by the tugboat onto the barge. This discrepancy means that previous research could not adequately explain how the specified pushing force is generated, or whether it can be achieved. Therefore, from the perspective of practical application, the pushing force assumption may be inappropriate.

The most direct approach involves establishing a numerical model for the collision and elastic deformation between the tugboats and the barge, but this substantially increases both model complexity and computational burdens. Accordingly, this paper remodels the multi-tugboats assisted pushing operation with specific assumptions to establish a simple and fast numerical simulation of barge pushing motion using the thrust of thrusters as control inputs.

2.2. Remodeling of the pushing operation

As illustrated in Fig. 5, the arrangement of the barge and the pushing tugboats studied in this paper is presented. The reference systems used in this paper is shown in Fig. 6, including the fixed reference system ($O_G X_G Y_G$), the body reference system of the barge ($O_B X_B Y_B$) and the body reference systems of the tugboats ($O_{T_i} X_{T_i} Y_{T_i}$). Based on the research by El-Hawary (2000), the low-frequency (LF) motion equations for the barge and the i th tugboat are given as follows:

$$\mathbf{M}^b \dot{\mathbf{v}}^b + \mathbf{C}_{RB}^b \mathbf{v}^b + \mathbf{D}_L^b \mathbf{v}^b = \boldsymbol{\tau}_{env}^b + \boldsymbol{\tau}_{push}^b \quad (1)$$

$$\mathbf{M}^{t,i} \dot{\mathbf{v}}^{t,i} + \mathbf{C}_{RB}^{t,i} \mathbf{v}^{t,i} + \mathbf{D}_L^{t,i} \mathbf{v}^{t,i} = \boldsymbol{\tau}_{env}^{t,i} + \boldsymbol{\tau}_{Thrust}^{t,i} + \begin{bmatrix} -T_{push}^{t,i} & 0 & 0 \end{bmatrix}^T \quad (2)$$

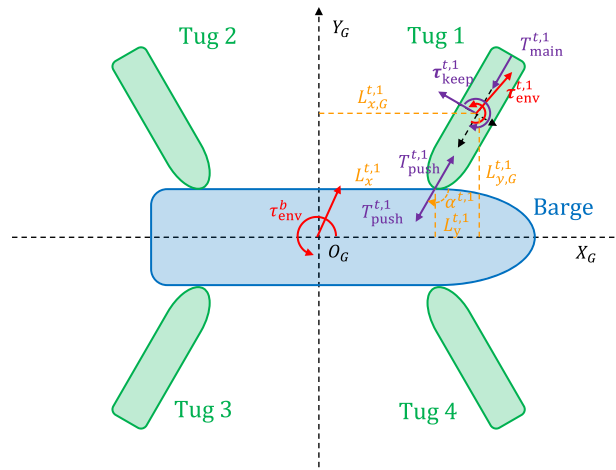


Fig. 5. Schematic diagram of the pushing operation system.

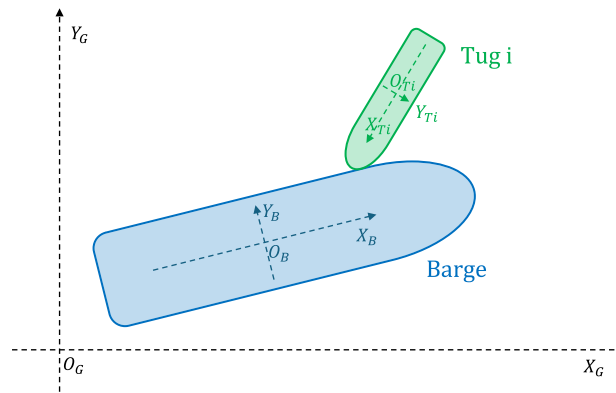


Fig. 6. The reference systems.

where $\mathbf{M}^b$ and $\mathbf{M}^{t,i}$ are the generalized inertial matrices of the barge and the tugboats, $\mathbf{C}_{RB}^b$ and $\mathbf{C}_{RB}^{t,i}$ are the Coriolis matrices, $\mathbf{D}_L^b$ and $\mathbf{D}_L^{t,i}$ are the linear damping matrices. $\mathbf{v}^b$ and $\mathbf{v}^{t,i}$ are the velocities of the barge and the tugboats. τ_{env}^b and $\tau_{env}^{t,i}$ are the environmental disturbances on the barge and the tugboats. τ_{push}^b is the resultant force and moment of the pushing force on the barge. $\tau_{Thruster}^{t,i}$ is the resultant force and moment on the tugboats, and is generated by the thrusters of the tugboats. $T_{push}^{t,i}$ is the pushing force generated by the tugboats. It is assumed that the pushing force generated by each tugboat is aligned with its longitudinal axis. Therefore, the tugboat experiences the pushing force of $[-T_{push}^{t,i} \ 0 \ 0]^T$ with its own reference system.

Both τ_{env}^b and $\tau_{env}^{t,i}$ are composed of second-order wave loads (τ_{wave2}), wind loads (τ_{wind}), and current loads ($\tau_{current}$). The values of τ_{wind} and $\tau_{current}$ are directly proportional to the square of wind velocity (V_w) and current velocity (V_c), while the value of τ_{wave2} is calculated using a quadratic transfer function (QTF). Specifically, τ_{wind} is defined as follows :

$$\tau_{wind} = \begin{bmatrix} C_{wx}(\gamma_{wr}) \\ C_{wy}(\gamma_{wr}) \\ C_{w\psi}(\gamma_{wr}) \end{bmatrix} \cdot U_{wr}^2 \quad (3)$$

Where C_{wx} , C_{wy} and $C_{w\psi}$ are the coefficients of the wind load, the γ_{wr} is the angle between ship heading and wind direction and the U_{wr} is the relative wind velocity of the vessel. Similarly, the current load is defined as follows:

$$\tau_{current} = \begin{bmatrix} C_{cx}(\gamma_{cr}) \\ C_{cy}(\gamma_{cr}) \\ C_{c\psi}(\gamma_{cr}) \end{bmatrix} \cdot U_{cr}^2 \quad (4)$$

Where C_{cx} , C_{cy} and $C_{c\psi}$ are the coefficients of the current load, the γ_{cr} is the angle between ship heading and current direction and the U_{cr} is the relative current velocity of the vessel. For τ_{wave2} , firstly, irregular waves are usually regarded as the linear superposition of several regular waves. One of the regular waves satisfies the following relation:

$$\zeta_i = A_i \cos(\omega_i t + \phi_i) \quad (5)$$

Where A_i is the wave amplitude and ω_i is the wave frequency, and the relationship between A_i and ω_i is determined by the wave spectrum used. ϕ_i is the initial phase and the ζ_i is the wave elevation. Therefore the τ_{wave2} is defined as:

$$\begin{aligned} \tau_{\text{wave2}} = & \sum_i \sum_j P_{ij} \cdot A_i A_j \cos [(\omega_i - \omega_j)t + \phi_i - \phi_j + \phi_{Pij}] \\ & + \sum_i \sum_j Q_{ij} \cdot A_i A_j \cos [(\omega_i - \omega_j)t + \phi_i - \phi_j + \phi_{Qij}] \end{aligned} \quad (6)$$

Where P_{ij} and Q_{ij} are the QTFs, ϕ_{Pij} and ϕ_{Qij} are the corresponding phases. The τ_{env}^b and $\tau_{\text{env}}^{t,i}$ are defined as:

$$\begin{aligned} \tau_{\text{env}}^b &= \tau_{\text{wave2}}^b + \tau_{\text{wind}}^b + \tau_{\text{current}}^b \\ \tau_{\text{env}}^{t,i} &= \tau_{\text{wave2}}^{t,i} + \tau_{\text{wind}}^{t,i} + \tau_{\text{current}}^{t,i} \end{aligned} \quad (7)$$

The influence of first-order wave loads on ship motion is typically calculated in the form of Response Amplitude Operators (RAOs) and superimposed with the results from the LF dynamic models, as η_{WF} in Fig. 4. Therefore, the related terms of first-order wave loads are not considered in Eq. (1) and (2).

τ_{push}^b is defined as:

$$\tau_{\text{push}}^b = \mathbf{B}_{\text{push}}^b(\alpha) \cdot \mathbf{T}_{\text{push}} \quad (8)$$

Where the vector α denotes the direction of pushing force generated by tugboats:

$$\alpha = [\alpha^{t,1} \quad \alpha^{t,2} \quad \dots \quad \alpha^{t,n}] \quad (9)$$

$\mathbf{B}_{\text{push}}^b$ signifies the direction matrix. The definition of $\mathbf{B}_{\text{push}}^b(\alpha)$ is:

$$\mathbf{B}_{\text{push}}^b(\alpha) = \begin{bmatrix} \cos \alpha^{t,1} & & \cos \alpha^{t,n} \\ \sin \alpha^{t,1} & \dots & \sin \alpha^{t,n} \\ L_x^{t,1} \sin \alpha^{t,1} - L_y^{t,1} \cos \alpha^{t,1} & \dots & L_x^{t,n} \sin \alpha^{t,n} - L_y^{t,n} \cos \alpha^{t,n} \end{bmatrix} \quad (10)$$

$(L_x^{t,i}, L_y^{t,i})$ denote the points of application for $T_{\text{push}}^{t,i}$. $\mathbf{T}_{\text{push}}$ represents the pushing force vector:

$$\mathbf{T}_{\text{push}} = [T_{\text{push}}^{t,1} \quad T_{\text{push}}^{t,2} \quad \dots \quad T_{\text{push}}^{t,n}]^T \quad (11)$$

This paper divides $\tau_{\text{thrust}}^{t,i}$ into $\tau_{\text{main}}^{t,i}$ and $\tau_{\text{keep}}^{t,i}$. The controller of the tugboats-barge system assigns the target value of $\tau_{\text{main}}^{t,i}$ for generating $T_{\text{push}}^{t,1}$ by main thrusters of the tugboats. $\tau_{\text{keep}}^{t,i}$ is determined by the controller of the tugboat itself and applied to generate the station keeping capability of the tugboat by side thrusters of the tugboats. Accordingly, Eq. (2) can be expressed as follows:

$$\mathbf{M}^{t,i} \dot{\mathbf{v}}^{t,i} + \mathbf{C}_{RB}^{t,i} \mathbf{v}^{t,i} + \mathbf{D}_L^{t,i} \mathbf{v}^{t,i} = \tau_{\text{env}}^{t,i} + \tau_{\text{main}}^{t,i} + \tau_{\text{keep}}^{t,i} + \begin{bmatrix} -T_{\text{push}}^{t,i} & 0 & 0 \end{bmatrix}^T \quad (12)$$

where $\tau_{\text{main}}^{t,i} = [T_{\text{main}}^{t,1} \quad 0 \quad 0]^T$ and $\tau_{\text{keep}}^{t,i} = [0 \quad F_{y,\text{keep}}^{t,i} \quad M_{\text{keep}}^{t,i}]^T$.

This paper assumes that each tugboat is equipped with an ideal controller capable of maintaining a fixed position relative to the barge, thereby enabling the solution of Eq. (1) and (2). Based on this assumption, $\mathbf{v}^b$ and $\mathbf{v}^{t,i}$ of the tugboat and the barge satisfy the following:

$$\mathbf{v}^{t,i} = \begin{bmatrix} \cos \alpha^{t,i} & \sin \alpha^{t,i} & 0 \\ -\sin \alpha^{t,i} & \cos \alpha^{t,i} & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{v}^b + \mathbf{r}^b \begin{bmatrix} \sin \alpha^{t,i} & -\cos \alpha^{t,i} & 0 \\ \cos \alpha^{t,i} & \sin \alpha^{t,i} & 0 \\ 0 & 0 & 0 \end{bmatrix} \boldsymbol{\eta}_{\text{body}}^{t,i} \quad (13)$$

$$\boldsymbol{\eta}_{\text{body}}^{t,i} = [L_x^{t,i} - L_G^{t,i} \cos \alpha^{t,i} \quad L_y^{t,i} - L_G^{t,i} \sin \alpha^{t,i} \quad \alpha^{t,i}]^T = [L_{x,G}^{t,i} \quad L_{y,G}^{t,i} \quad \alpha^{t,i}]^T \quad (14)$$

where $L_G^{t,i}$ refers to the distance from the center of gravity of the tugboat to its bow. Therefore, $\mathbf{v}^b$ and $\mathbf{v}^{t,i}$ of the tugboat and the barge satisfy the following:

$$\mathbf{v}^{t,i} = \begin{bmatrix} \cos \alpha^{t,i} & \sin \alpha^{t,i} & 0 \\ -\sin \alpha^{t,i} & \cos \alpha^{t,i} & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{v}^b + \mathbf{r}^b \begin{bmatrix} \sin \alpha^{t,i} & -\cos \alpha^{t,i} & 0 \\ \cos \alpha^{t,i} & \sin \alpha^{t,i} & 0 \\ 0 & 0 & 0 \end{bmatrix} \boldsymbol{\eta}_{\text{body}}^{t,i} \quad (15)$$

The complete forms of Eq. (1) and (2) are as follows:

$$\begin{aligned} & \begin{bmatrix} m^b - X_u^b & 0 & 0 \\ 0 & m^b - Y_v^b & m^b x_G^b - Y_r^b \\ 0 & m^b x_G^b - N_v^b & m^b x_G^b - N_r^b \end{bmatrix} \cdot \begin{bmatrix} \dot{u}^b \\ \dot{v}^b \\ \dot{r}^b \end{bmatrix} + \begin{bmatrix} 0 & 0 & -m^b (v^b + x_G^b r^b) \\ 0 & 0 & m^b u^b \\ m^b (v^b + x_G^b r^b) & -m^b u^b & 0 \end{bmatrix} \cdot \begin{bmatrix} u^b \\ v^b \\ r^b \end{bmatrix} \\ & - \begin{bmatrix} X_u^b & 0 & 0 \\ 0 & Y_v^b & Y_r^b \\ 0 & N_v^b & N_r^b \end{bmatrix} \cdot \begin{bmatrix} u^b \\ v^b \\ r^b \end{bmatrix} = \begin{bmatrix} F_{x,\text{env}}^{t,i} \\ F_{y,\text{env}}^{t,i} \\ M_{\text{env}}^{t,i} \end{bmatrix} + \begin{bmatrix} \cos \alpha^{t,1} & & \cos \alpha^{t,n} \\ \sin \alpha^{t,1} & \dots & \sin \alpha^{t,n} \\ L_x^{t,1} \sin \alpha^{t,1} - L_y^{t,1} \cos \alpha^{t,1} & \dots & L_x^{t,n} \sin \alpha^{t,n} - L_y^{t,n} \cos \alpha^{t,n} \end{bmatrix} \cdot \begin{bmatrix} T_{\text{push}}^{t,1} \\ \dots \\ T_{\text{push}}^{t,n} \end{bmatrix} \end{aligned} \quad (16)$$

$$\begin{aligned} & \begin{bmatrix} m^{t,i} - X_u^{t,i} & 0 & 0 \\ 0 & m^{t,i} - Y_v^{t,i} & m^{t,i} x_G^{t,i} - Y_r^{t,i} \\ 0 & m^{t,i} x_G^{t,i} - N_v^{t,i} & m^{t,i} x_G^{t,i} - N_r^{t,i} \end{bmatrix} \cdot \begin{bmatrix} \dot{u}^{t,i} \\ \dot{v}^{t,i} \\ \dot{r}^{t,i} \end{bmatrix} + \begin{bmatrix} 0 & 0 & -m^{t,i} (v^{t,i} + x_G^{t,i} r^{t,i}) \\ 0 & 0 & m^{t,i} u^{t,i} \\ m^{t,i} (v^{t,i} + x_G^{t,i} r^{t,i}) & -m^{t,i} u^{t,i} & 0 \end{bmatrix} \cdot \begin{bmatrix} u^{t,i} \\ v^{t,i} \\ r^{t,i} \end{bmatrix} \\ & - \begin{bmatrix} X_u^{t,i} & 0 & 0 \\ 0 & Y_v^{t,i} & Y_r^{t,i} \\ 0 & N_v^{t,i} & N_r^{t,i} \end{bmatrix} \cdot \begin{bmatrix} u^{t,i} \\ v^{t,i} \\ r^{t,i} \end{bmatrix} = \begin{bmatrix} F_{x,\text{env}}^{t,i} \\ F_{y,\text{env}}^{t,i} \\ M_{\text{env}}^{t,i} \end{bmatrix} + \begin{bmatrix} T_{\text{main}}^{t,i} \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ F_{y,\text{keep}}^{t,i} \\ M_{\text{keep}}^{t,i} \end{bmatrix} + \begin{bmatrix} -T_{\text{push}}^{t,i} \\ 0 \\ 0 \end{bmatrix} \end{aligned} \quad (17)$$

Table 1
Parameters of the barge and tugboats.

Barge			Tug		
Displacement	kg	1.39e8	Displacement	kg	8.92e6
Draft	m	17.25	Draft	m	6.90
Beam	m	50.00	Beam	m	20.00
Length	m	200.00	Length	m	80.00

Table 2
Parameters of the tugboats and their thrust.

	$L_x^{t,i}$ (m)	$L_y^{t,i}$ (m)	$L_G^{t,i}$ (m)	$\alpha^{t,i}$ (deg)	$T_{main}^{t,i}$ (KN)	$\Delta T_{main}^{t,i}$ (KN/s)
Tugboat 1	60	25	45	-120	[200, 2500]	[-150, 150]
Tugboat 2	-60	25	45	-60	[200, 2500]	[-150, 150]
Tugboat 3	-60	-25	45	60	[200, 2500]	[-150, 150]
Tugboat 4	60	-25	45	120	[200, 2500]	[-150, 150]

Let $M_{11}^{t,i} = m^{t,i} - X_u^{t,i}$, $C_{13}^{t,i} = -m^{t,i} (v^{t,i} + x_G^{t,i} r^{t,i})$, $D_{11}^{t,i} = -X_u^{t,i}$, According to Eq. (17), $T_{push}^{t,i}$ is defined as:

$$\begin{aligned}
 T_{push}^{t,i} &= -M_{11}^{t,i} \dot{u}^{t,i} - C_{13}^{t,i} r^{t,i} - D_{11}^{t,i} u^{t,i} + F_{x,env}^{t,i} + T_{main}^{t,i} \\
 &= -M_{11}^{t,i} \left\{ \dot{u}^b \cos \alpha^{t,i} + v^b \sin \alpha^{t,i} + r^b \left[L_{x,G}^{t,i} \sin \alpha^{t,i} - L_{y,G}^{t,i} \cos \alpha^{t,i} \right] \right\} - C_{13}^{t,i} r^b \\
 &\quad - D_{11}^{t,i} \left\{ u^b \cos \alpha^{t,i} + v^b \sin \alpha^{t,i} + r^b \left[L_{x,G}^{t,i} \sin \alpha^{t,i} - L_{y,G}^{t,i} \cos \alpha^{t,i} \right] \right\} + F_{x,env}^{t,i} + T_{main}^{t,i}
 \end{aligned} \quad (18)$$

Therefore:

$$B_{push}^b(\alpha) \cdot T_{push} = \sum_{i=1}^n -\tilde{M}^{t,i} \dot{v}_b - \tilde{D}^{t,i} v_b - \tilde{C}^{t,i} v_b + \tilde{\tau}_{env}^{t,i} + \tilde{\tau}_{main}^{t,i} \quad (19)$$

where:

$$\begin{aligned}
 \tilde{M}^{t,i} &= M_{11}^{t,i} \begin{bmatrix} \cos \alpha^{t,i} \\ \sin \alpha^{t,i} \\ L_x^{t,i} \sin \alpha^{t,i} - L_y^{t,i} \cos \alpha^{t,i} \end{bmatrix} \cdot \begin{bmatrix} \cos \alpha^{t,i} & \sin \alpha^{t,i} & L_{x,G}^{t,i} \sin \alpha^{t,i} - L_{y,G}^{t,i} \cos \alpha^{t,i} \end{bmatrix} \\
 \tilde{D}^{t,i} &= D_{11}^{t,i} \begin{bmatrix} \cos \alpha^{t,i} \\ \sin \alpha^{t,i} \\ L_x^{t,i} \sin \alpha^{t,i} - L_y^{t,i} \cos \alpha^{t,i} \end{bmatrix} \cdot \begin{bmatrix} \cos \alpha^{t,i} & \sin \alpha^{t,i} & L_{x,G}^{t,i} \sin \alpha^{t,i} - L_{y,G}^{t,i} \cos \alpha^{t,i} \end{bmatrix} \\
 \tilde{C}^{t,i} &= C_{13}^{t,i} \begin{bmatrix} \cos \alpha^{t,i} \\ \sin \alpha^{t,i} \\ L_x^{t,i} \sin \alpha^{t,i} - L_y^{t,i} \cos \alpha^{t,i} \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \\
 \tilde{\tau}_{env}^{t,i} &= F_{x,env}^{t,i} \begin{bmatrix} \cos \alpha^{t,i} \\ \sin \alpha^{t,i} \\ L_x^{t,i} \sin \alpha^{t,i} - L_y^{t,i} \cos \alpha^{t,i} \end{bmatrix}, \quad \tilde{\tau}_{main}^{t,i} = T_{main}^{t,i} \begin{bmatrix} \cos \alpha^{t,i} \\ \sin \alpha^{t,i} \\ L_x^{t,i} \sin \alpha^{t,i} - L_y^{t,i} \cos \alpha^{t,i} \end{bmatrix}
 \end{aligned} \quad (20)$$

Consequently, the dynamic equation of the barge can be rewritten as:

$$M^{re} \dot{v}^b + C_{RB}^{re} v^b + D_L^{re} v^b = \tau_{env}^{re} + \tau_{main}^b \quad (21)$$

where $M^{re} = M^b + \sum_{i=1}^n \tilde{M}^{t,i}$, $C_{RB}^{re} = C_{RB}^b + \sum_{i=1}^n \tilde{C}^{t,i}$, $D_L^{re} = D_L^b + \sum_{i=1}^n \tilde{D}^{t,i}$, $\tau_{env}^{re} = \tau_{env}^b + \sum_{i=1}^n \tilde{\tau}_{env}^{t,i}$, $\tau_{main}^b = \sum_{i=1}^n \tilde{\tau}_{main}^{t,i}$.

The main parameters of the tugboats and barge used in this study are listed in Table 1, and identical tugboats are used. The parameters of the thrust are listed in Table 2. The main thrusters thrust $T_{main}^{t,i}$ for each tugboat is provided by a pair of thrusters. All the thrusters are fixed and ignore the relevant interactions.

Therefore:

$$\sum_{i=1}^n \tilde{M}^{t,i} = M_{11}^{t,i} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6.23e3 \end{bmatrix} \quad (22)$$

This demonstrates that the reconstructed generalized mass matrix is nonsingular.

2.3. The sub-modules of present mathematical model

For present model, the simulation procedure can be established as shown in Fig. 7.

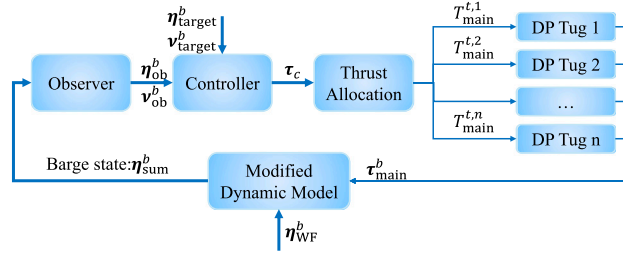


Fig. 7. The simulation procedure of the pushing operation with the proposed model.

The observer for the barge is employed to estimate its LF motion. This study uses the nonlinear passive observer (NPO) proposed by Fossen and Strand (1999), and the structure is outlined as follows:

$$\begin{aligned}
 \tilde{\mathbf{y}} &= \boldsymbol{\eta}_{\text{sum}}^b - \hat{\mathbf{y}} \\
 \dot{\tilde{\boldsymbol{\xi}}} &= \mathbf{A}_W \tilde{\boldsymbol{\xi}} + \mathbf{K}_1 \tilde{\mathbf{y}} \\
 \dot{\hat{\boldsymbol{\eta}}} &= \mathbf{R}(\hat{\boldsymbol{\psi}}) \hat{\mathbf{v}} + \mathbf{K}_2 \tilde{\mathbf{y}} \\
 \dot{\hat{\mathbf{b}}} &= \mathbf{T}^{-1} \hat{\mathbf{b}} + \mathbf{K}_3 \tilde{\mathbf{y}} \\
 \mathbf{M} \dot{\hat{\mathbf{v}}} &= -\mathbf{D} \hat{\mathbf{v}} + \mathbf{R}^T(\hat{\boldsymbol{\psi}}) \hat{\mathbf{b}} + \boldsymbol{\tau}_c + \mathbf{K}_4 \mathbf{R}^T(\hat{\boldsymbol{\psi}}) \tilde{\mathbf{y}} \\
 \dot{\hat{\mathbf{y}}} &= \hat{\boldsymbol{\eta}} + \mathbf{C}_W \tilde{\boldsymbol{\xi}}
 \end{aligned} \tag{23}$$

where $\mathbf{K}_1, \mathbf{K}_2, \mathbf{K}_3, \mathbf{K}_4, \mathbf{A}_W, \mathbf{C}_W$ are gain matrices. $\hat{\boldsymbol{\eta}}$ represents the estimated LF motion in the global coordinate system. $\hat{\mathbf{b}}$ denotes the estimated slowly varying environmental disturbances and unmodeled dynamics. $\tilde{\boldsymbol{\xi}}$ stands for the estimated wave-frequency (WF) motion of the vessel.

A sliding mode controller (SMC) is utilized to regulate the motion of the barge. The sliding surface is defined as follows:

$$s_{sf} = \dot{e} + \boldsymbol{\Lambda} e + \mathbf{K}_i \cdot \int e dt \tag{24}$$

where $e = \mathbf{R}^T(\boldsymbol{\psi}) (\boldsymbol{\eta}_{ob}^b - \boldsymbol{\eta}_{\text{target}}^b)$ represents the positioning errors, $\boldsymbol{\Lambda}$ is the gain matrix of the sliding surface and $\mathbf{K}_i$ is the gain matrix for the integral term. The output $\boldsymbol{\tau}_c$ of the SMC can be defined as:

$$\boldsymbol{\tau}_c = \boldsymbol{\tau}_{eq} + \boldsymbol{\tau}_{sw} \tag{25}$$

The equivalent control force $\boldsymbol{\tau}_{eq}$ is computed based on the known dynamic parameters:

$$\boldsymbol{\tau}_{eq} = -\mathbf{M} \boldsymbol{\Lambda} \dot{e} + \mathbf{D}_L \mathbf{v}_{ob}^b + \mathbf{C}_{RB}(\mathbf{v}_{ob}^b) \mathbf{v}_{ob}^b \tag{26}$$

The switching control force is defined as follows:

$$\boldsymbol{\tau}_{sw} = -\mathbf{K}_{sw} \cdot \min \left[\max \left(\frac{s_{sf}}{\phi}, -1 \right), 1 \right] \tag{27}$$

where $\mathbf{K}_{sw}$ is the switching gain matrix, and ϕ is the boundary layer thickness of the sliding surface.

Thrust allocation is realized by solving the following quadratic modification problem, which is proposed by Johansen et al. (2004):

$$\begin{aligned}
 \min J(\mathbf{T}_{\text{main}}, s) &= \mathbf{T}_{\text{main}}^T \mathbf{W} \mathbf{T}_{\text{main}} + s^T \mathbf{Q} s \\
 \text{s.t. } s + \mathbf{B}_{\text{main}}^b(\boldsymbol{\alpha}) \mathbf{T}_{\text{main}} &= \boldsymbol{\tau}_c \\
 \Delta \mathbf{T}_{\text{min}} &\leq \mathbf{T}_{\text{main}} - \mathbf{T}_{\text{main}0} \leq \Delta \mathbf{T}_{\text{max}} \\
 \mathbf{T}_{\text{min}} &\leq \mathbf{T}_{\text{main}} \leq \mathbf{T}_{\text{max}}
 \end{aligned} \tag{28}$$

where s is a slack variable, $\mathbf{W}$ and $\mathbf{Q}$ are weight matrices for thrust and slack variables. $\mathbf{T}_{\text{main}0}$ is the $\mathbf{T}_{\text{main}}$ of the last step. $\Delta \mathbf{T}_{\text{min}}$ and $\Delta \mathbf{T}_{\text{max}}$ denote the lower and upper bounds for thrust change rate, while $\mathbf{T}_{\text{min}}$ and $\mathbf{T}_{\text{max}}$ are the lower and upper bounds for thrust. $\mathbf{B}_{\text{main}}^b, \mathbf{T}_{\text{main}}$ and $\mathbf{B}_{\text{push}}^b, \mathbf{T}_{\text{push}}$ have similar definitions.

3. Comparison of numerical simulation results with proposed and conventional mathematical models

In some relevant studies, further simplification of the numerical simulations has been performed by neglecting the WF motion of the barge. On this basis, the observer is replaced with an ideal observer which can provide accurate LF motions for the controller. Thus, the simulation procedures are as shown in Figs. 8 and 9.

A station keeping scenario of a barge is simulated with the same environmental disturbances as well as controller parameters for both conventional and proposed model as listed in Tables 3 and 4. The numerical simulation settings are shown in Table 5. This paper will analyze the differences between the conventional model and the proposed model by the simulation results of the motion, the resultant forces and moments and the pushing force and main thruster thrust.

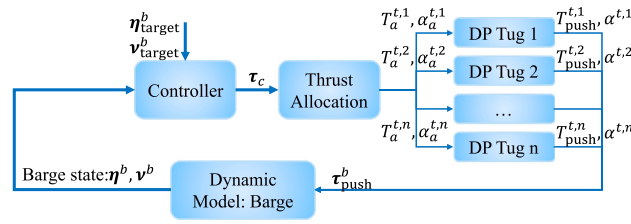


Fig. 8. The simulation procedure of the pushing operation with the conventional model and an ideal observer.

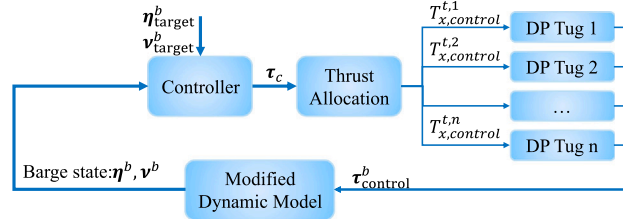


Fig. 9. The simulation procedure of the pushing operation with the proposed model and an ideal observer.

Table 3
Parameters of environmental disturbances.

H_s (m)	T_p (s)	Spectrum	V_w (m/s)	V_c (knots)	Direction (deg)
2.5	8.0	JONSWAP	10.0	1.0	90,135,180

Table 4
Parameters of the SMC.

Direction (deg)	Λ	K_{sw}	ϕ	K_i
90	diag ([1, 1, 1])	diag ((1e4, 2e6, 2e7))	diag ([0.05, 0.05, 0.01])	diag ((1e-3, 1e-3, 1e-2))
135		diag ([5e5, 2e6, 5e7])		
180		diag ([5e5, 1e5, 1e7])		

Table 5
Parameters of the simulation settings.

Simulation step (s)	Control step (s)	Simulation duration (s)	Solver
0.5	1.0	5400	Runge-Kutta 4

Table 6
The statistical results of the barge motion with the conventional model and an ideal observer.

Direction		Mean	Max.	Min.	Std.	IAE (LF)
90 deg	x (m)	0.002	0.093	-0.081	0.024	11.897
	y (m)	0.003	2.069	-2.068	0.575	53.278
	ψ (deg)	-0.001	0.139	-0.144	0.042	14.229
135 deg	x (m)	-0.002	0.876	-0.888	0.236	22.372
	y (m)	0.001	0.972	-0.941	0.261	11.133
	ψ (deg)	0.001	0.689	-0.708	0.195	14.218
180 deg	x (m)	-0.005	0.761	-0.814	0.219	31.412
	y (m)	0.000	0.000	0.000	0.000	0.000
	ψ (deg)	0.000	0.000	0.000	0.000	0.000

3.1. The simulation results of motion response

In this paper, the following types of numerical simulations are used to test the difference between the proposed model and the traditional model: the tugboats assist the barge to resist environmental disturbances and maintain the position. The tugboat and barge are started together with the environmental disturbance at the target position. This paper retains the numerical simulation results of the 5400 s when the system reaches a stable state.

The motion results of the barge with environmental disturbances at 135deg are shown in Fig. 10 as an example. The statistical results at each direction are summarized in Tables 6 and 7. The integral of absolute error (IAE) corresponding to the LF positioning errors e is defined as follows:

$$\text{IAE} = \int_0^t |e| dt \quad (29)$$

From the results, it can be seen that, with appropriate parameter selection, the modification of the mathematical model does not significantly affect the motion results of the barge. The average, maximum, minimum, and standard deviation values are similar and the IAEs for some cases

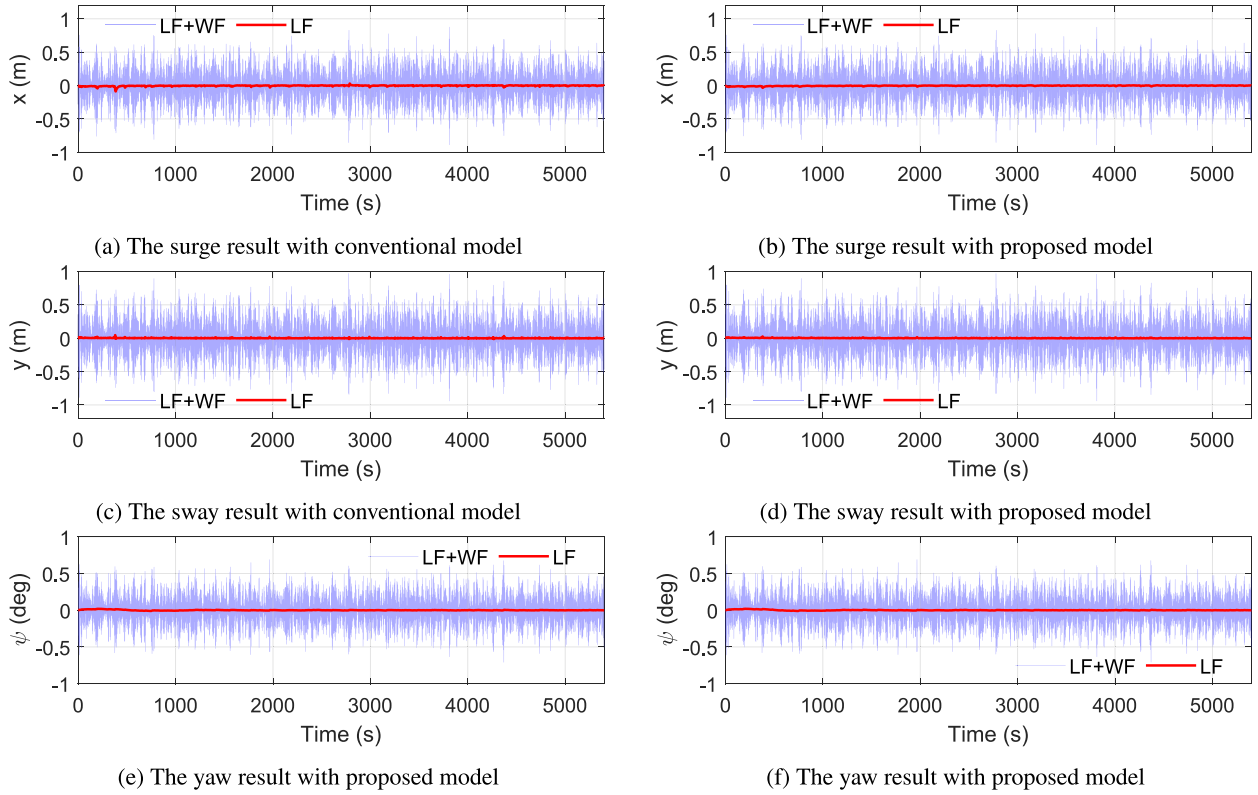


Fig. 10. The motion results of the barge with an ideal observer (135deg).

Table 7

The statistical results of the barge motion with the proposed model and an ideal observer.

Direction		Mean	Max.	Min.	Std.	IAE (LF)
90 deg	x (m)	0.001	0.092	-0.081	0.024	4.899
	y (m)	0.003	2.071	-2.075	0.575	32.219
	ψ (deg)	-0.001	0.141	-0.144	0.042	13.637
135 deg	x (m)	-0.003	0.875	-0.889	0.236	19.380
	y (m)	0.001	0.969	-0.939	0.261	12.405
	ψ (deg)	0.001	0.690	-0.708	0.195	14.806
180 deg	x (m)	-0.003	0.760	-0.815	0.219	24.576
	y (m)	0.000	0.000	0.000	0.000	0.000
	ψ (deg)	0.000	0.000	0.000	0.000	0.000

of proposed model are even less than those of conventional model. This is because while the control accuracy of the SMC is influenced by the model accuracy, the chattering of the controller output is also affected by K_{sw} . The more accurate the model, the higher the control precision is; the closer K_{sw} is to the true disturbances bound, the smaller the output chattering is. Without altering the input parameters, the model accuracy inside the SMC decreases after modification, but the actual disturbances of the system increases due to additional tugboats-induced disturbances, expanding the bound and bringing it closer to K_{sw} . Consequently, the control precision of both models is comparable.

3.2. The simulation results of the resultant forces and moments

The statistical results of τ_{push}^b and τ_{main}^b are illustrated in Fig. 11. The results of τ_{push}^b and τ_{main}^b validate the rationality of the model modification: with similar positioning performance, the τ_{push}^b with conventional and proposed model are nearly identical with the same environmental disturbances, both serving to balance LF environmental disturbances acting on the barge; meanwhile, the τ_{main}^b are bigger than the τ_{push}^b . This is because the τ_{main}^b helps the barge as well as the tugboats themselves resist environmental disturbances.

In order to validate this conclusion and demonstrate the accuracy of the proposed model, Tables 8~10 present the mean values of environmental disturbances acting on the barge and tugboats, as well as the combined mean values of environmental disturbances on the barge and those transmitted from the tugboats to the barge. These mean values correspond respectively to τ_{env}^b , $\tau_{env}^{t,i}$ and τ_{env}^{re} in Eq. (1), (2) and (21). The final results indicate that the mean value of τ_{env}^b is essentially the additive inverse of the mean value of τ_{push}^b , and the mean value of τ_{env}^{re} is essentially the additive inverse of the mean value of τ_{main}^b , thereby providing strong validation for the aforementioned conclusion.

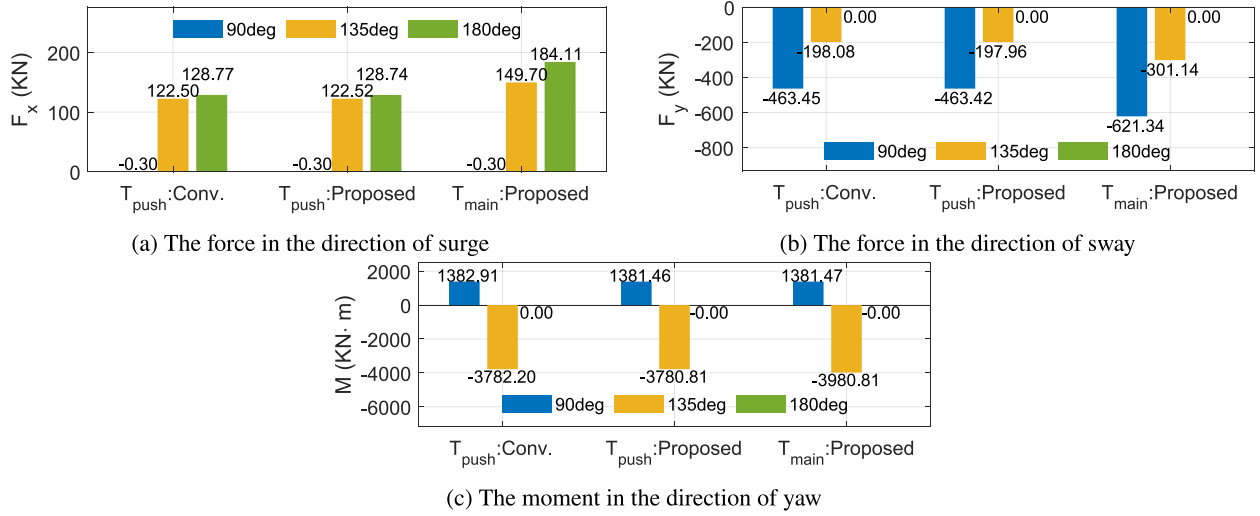


Fig. 11. The mean values of τ_{push}^b and τ_{main}^b with an ideal observer.

Table 8

The mean values of environmental disturbances (90 deg).

	Barge	tugboat 1	tugboat 2	tugboat 3	tugboat 4	Sum
$F_{x,env}$ (KN)	0.30	-44.87	-44.87	46.32	46.32	0.30
$F_{y,env}$ (KN)	463.85	-39.18	39.18	34.15	-34.15	621.79
M_{env} (KN m)	-1381.72	-608.18	608.17	-612.46	612.45	-1381.73

Table 9

The mean values of environmental disturbances (135 deg).

	Barge	tugboat 1	tugboat 2	tugboat 3	tugboat 4	Sum
$F_{x,env}$ (KN)	-122.57	-17.27	-41.92	15.12	44.84	-149.76
$F_{y,env}$ (KN)	198.07	-114.48	-19.03	116.52	16.10	301.25
M_{env} (KN m)	3784.45	163.99	-331.94	-137.83	-378.03	3984.54

Table 10

The mean values of environmental disturbances (180 deg).

	Barge	tugboat 1	tugboat 2	tugboat 3	tugboat 4	Sum
$F_{x,env}$ (KN)	-128.80	27.17	-28.20	-28.20	27.17	-184.16
$F_{y,env}$ (KN)	0.00	-84.34	-82.87	82.87	84.34	0.00
M_{env} (KN m)	0.00	385.86	-515.11	515.11	-385.86	0.00

3.3. The simulation results of the pushing force and main thruster thrust

The statistical results of $T_{push}^{t,i}$ and $T_{main}^{t,i}$ are illustrated in Figs. 12 – 13. Firstly, it can be observed that there is a clear difference between the $T_{push}^{t,i}$ and $T_{main}^{t,i}$, corroborating previous equations. The differences in $T_{push}^{t,i}$ with proposed and conventional model are mainly influenced by two factors: first, the optimization objective for thrust allocation is the sum of squares of the $T_{main}^{t,i}$ rather than that of the $T_{push}^{t,i}$; second, the outputs of the SMC are different because of different mathematical models. Because of the significant differences between pushing force and main thruster thrust, the proposed model should be used when focusing on controller output and thrusters response.

When analyzing the response of the thrusters, it should be noted that the performance of the thrusters not only limits the upper and lower bounds of the thrust values but also imposes constraints on the upper and lower bounds of the rate of thrust change per unit time. Therefore, this paper further discusses the rate of thrust change. During this analysis, it is discovered that there are instances where the rate of thrust change reaches the performance limit of the thruster, i.e., thruster saturation. The thrusters on Tugboat 2 exhibit the highest frequency of this condition. The corresponding numerical simulation results are shown in Fig. 14, where red dots indicate the values of $\Delta T_{push}^{t,i}$ at the moments of thruster saturation. The statistics results of thruster saturation are shown in Table 11. The statistics also record the frequency of saturation occurrences and the maximum loss rate of the $\Delta T_{main}^{t,i}$, where the loss rate $L_{\Delta T}$ is defined as follows:

$$L_{\Delta T} = \frac{|\Delta T_{main}^{sat} - \Delta T_{push}^{sat}|}{|\Delta T_{push}^{sat}|} \cdot 100\% \quad (30)$$

where ΔT_{main}^{sat} , ΔT_{push}^{sat} represent $\Delta T_{main}^{t,i}$, $\Delta T_{push}^{t,i}$ when thrusters saturation. $L_{\Delta T}$ indicates that with environmental disturbances, the tugboat may sometimes require a bigger $\Delta T_{main}^{t,i}$ to achieve the specified $\Delta T_{push}^{t,i}$. Therefore, when performing numerical simulations with the conventional model, the $\Delta T_{push}^{t,i}$ should be more conservative than those for the main thrusters of the tugboats — for example, scaling down according to the maximum $L_{\Delta T}$.

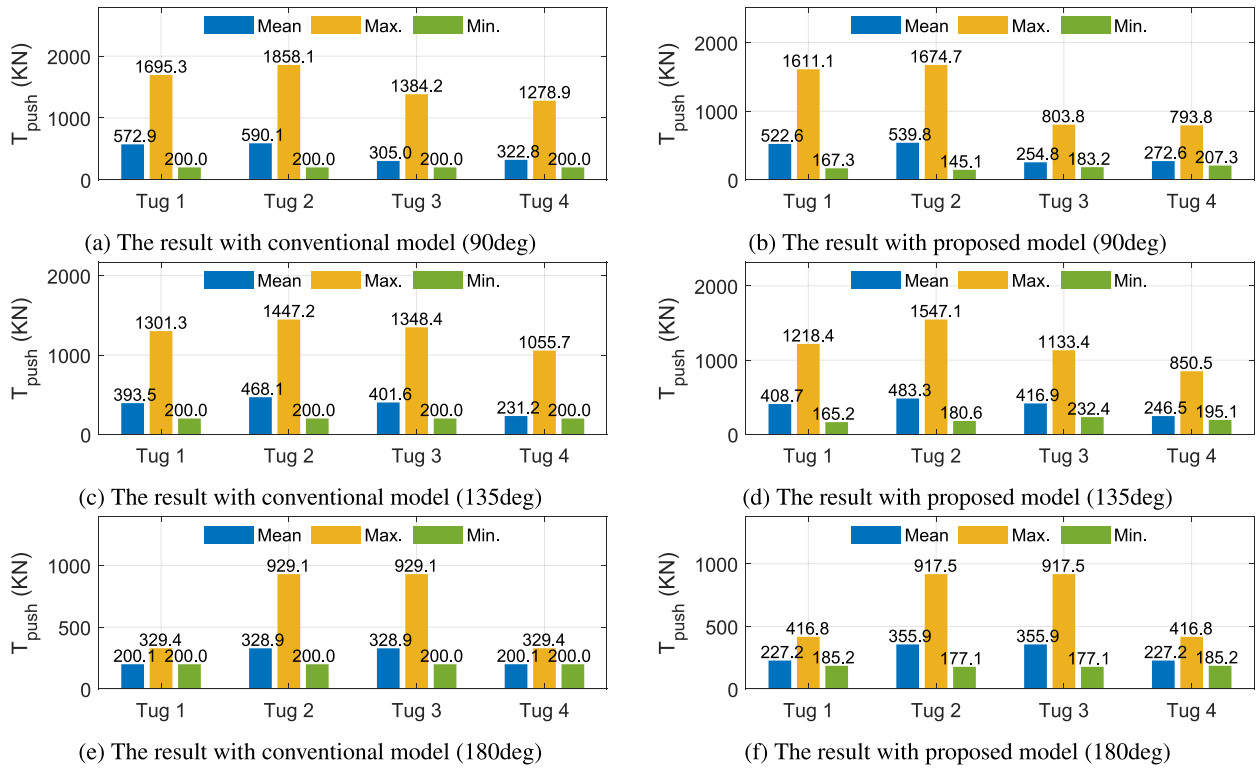


Fig. 12. The statistical results of $T_{push}^{f,i}$ with an ideal observer.

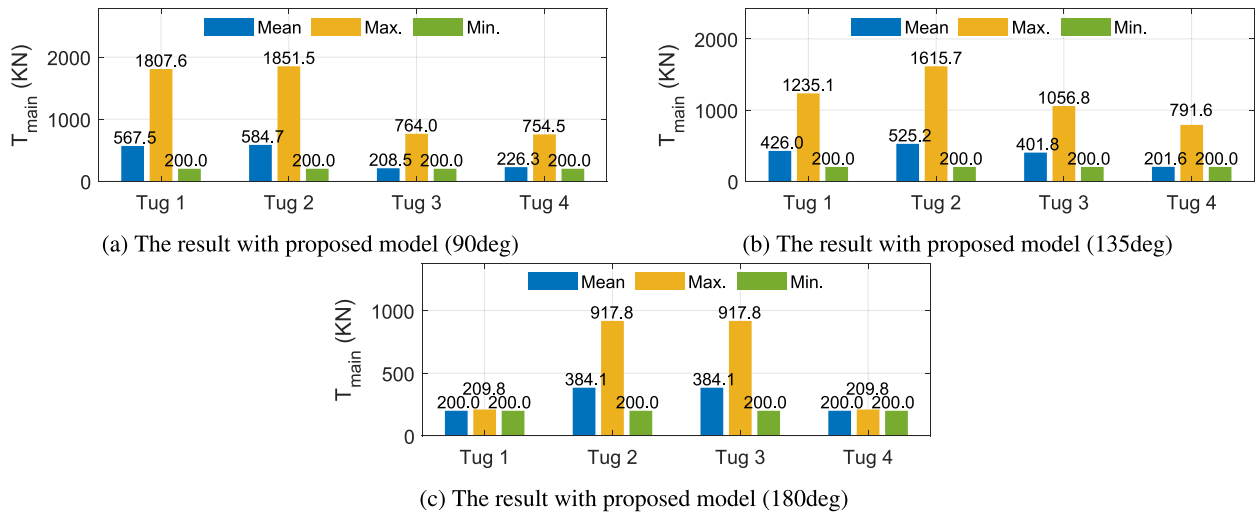


Fig. 13. The statistical results of $T_{main}^{f,i}$ with an ideal observer.

Table 11

The statistical results of thruster saturation with an ideal observer.

tugboat ID	90 deg		135 deg		180d eg	
	Count	Max.: L_{AT} (%)	Count	Max.: L_{AT} (%)	Count	Max.: L_{AT} (%)
1	110	28.04	6	3.77	0	0.00
2	404	28.12	158	22.48	1	10.68
3	27	16.30	1	10.61	0	-7.88
4	9	7.01	6	9.75	0	0.00

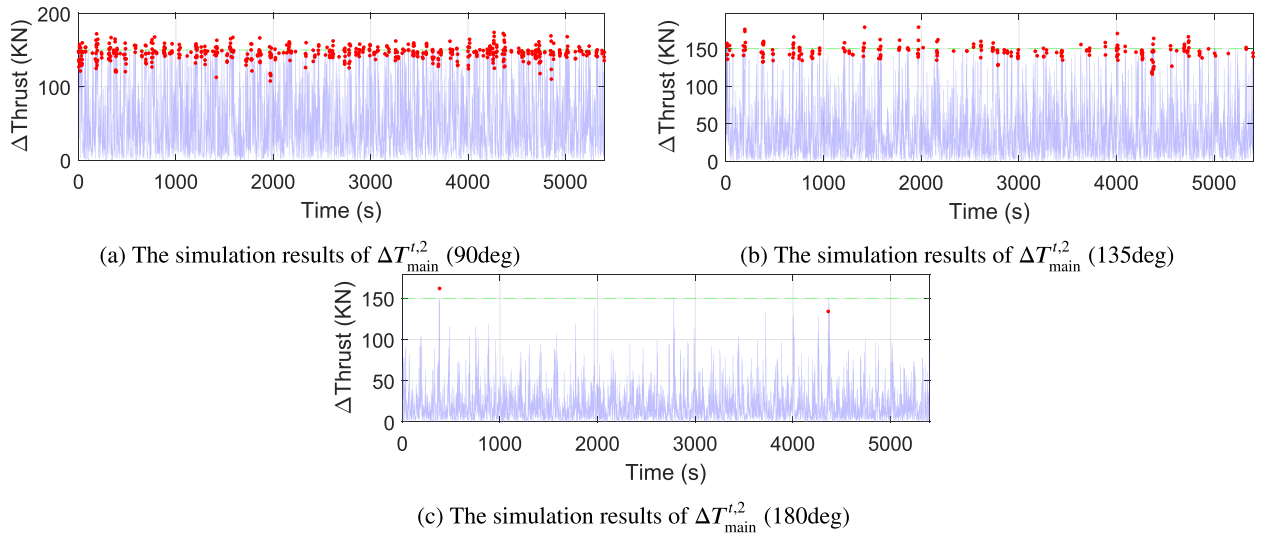


Fig. 14. The simulation results of $\Delta T_{\text{main}}^{r,2}$ with an ideal observer.

Table 12

Parameters of the SMC.

Direction (deg)	Λ	K_{sw}	ϕ	K_i
90		diag ([5e5, 5e6, 1e9])		
135	diag ([1, 1, 1])	diag ([1e6, 2e6, 2e8])	diag ([0.1, 0.1, 0.025])	diag ([1e-3, 1e-3, 1e-2])
180		diag ([2e6, 1e6, 1e8])		

Table 13

The statistical results of the barge motion with the proposed model and a real observer.

Direction		Mean	Max.	Min.	Std.	IAE (LF)
90 deg	x (m)	-0.001	0.096	-0.106	0.025	29.306
	y (m)	0.027	2.144	-2.137	0.584	476.552
	ψ (deg)	-0.010	0.196	-0.191	0.054	156.835
135 deg	x (m)	0.037	0.986	-0.904	0.255	434.997
	y (m)	0.009	0.945	-0.947	0.262	139.731
	ψ (deg)	0.006	0.687	-0.747	0.195	79.702
180 deg	x (m)	0.000	0.858	-0.794	0.227	264.923
	y (m)	0.000	0.000	0.000	0.000	0.012
	ψ (deg)	0.000	0.000	0.000	0.000	0.004

It is evident that thruster saturation primarily occurs in units with thrust vectors opposite to environmental disturbances, with the highest observed loss rate nearing 30%. This implies that if the conventional model is used for calculations, the $\Delta T_{\text{main}}^{r,i}$ should be reduced from $[-150, 150]$ kN/s to $[-105, 105]$ kN/s. Notably, these findings are grounded on simulations using an ideal observer. If the η_{WF} and a real observer are considered, the controller chattering will increase significantly, making thruster saturation more frequently.

4. The discussion on thruster saturation during the pushing operation with a real observer

In this section, the numerical simulations with a real observer are first presented, with the aim of demonstrating the impact of the real observer on the simulation results while determining the corresponding loss rate. Subsequently, based on the obtained loss rate, its influence on the numerical simulation results of the conventional model is discussed.

4.1. Thruster saturation with a real observer

Considering WF motion, the NPO (Eq. (23)) is used to estimate the LF motion of the barge and serves as the controller input. The parameters of the SMC are listed in Table 12. Subjected to the same environmental disturbances, the simulation result with environmental disturbances in 135deg is shown in Fig. 15 and the statistical results for each direction are detailed in Table 13. It can be seen that when employing the real observer, the mean value of the barge motion remains close to zero, but its IAE increases significantly. This is attributed to both magnitude and phase discrepancies between the observed and actual LF motions, leading to persistent oscillations near zero and heightened controller chattering.

It can be seen that the mean value of the τ_{main}^b in Table 14 is similar to the results presented in Fig. 11, but the statistical results of $T_{\text{main}}^{r,i}$ in Fig. 16 is noticeably higher compared to that in Fig. 13. This occurs because observer errors necessitate a larger K_{sw} for the SMC, which in turn amplifies the chattering in the control output. Consequently, Table 15 shows that thruster saturation becomes more frequently. This further demonstrates that, for the conventional model, using the parameter of the tugboat thruster setting $\Delta T_{\text{main}}^{r,i}$ is inappropriate.

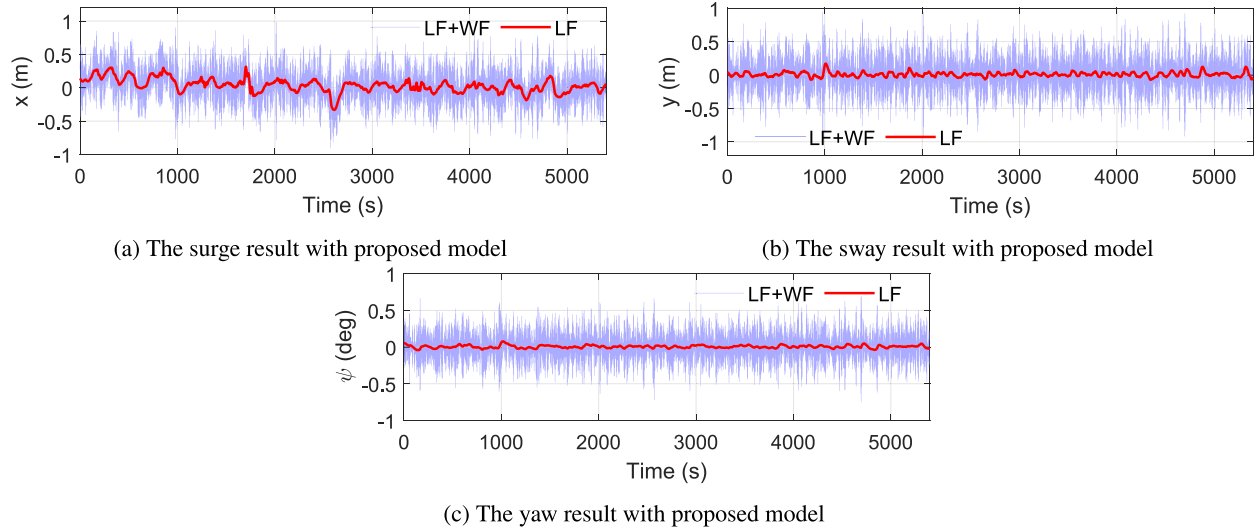


Fig. 15. The motion results of the barge with a real observer (135deg).

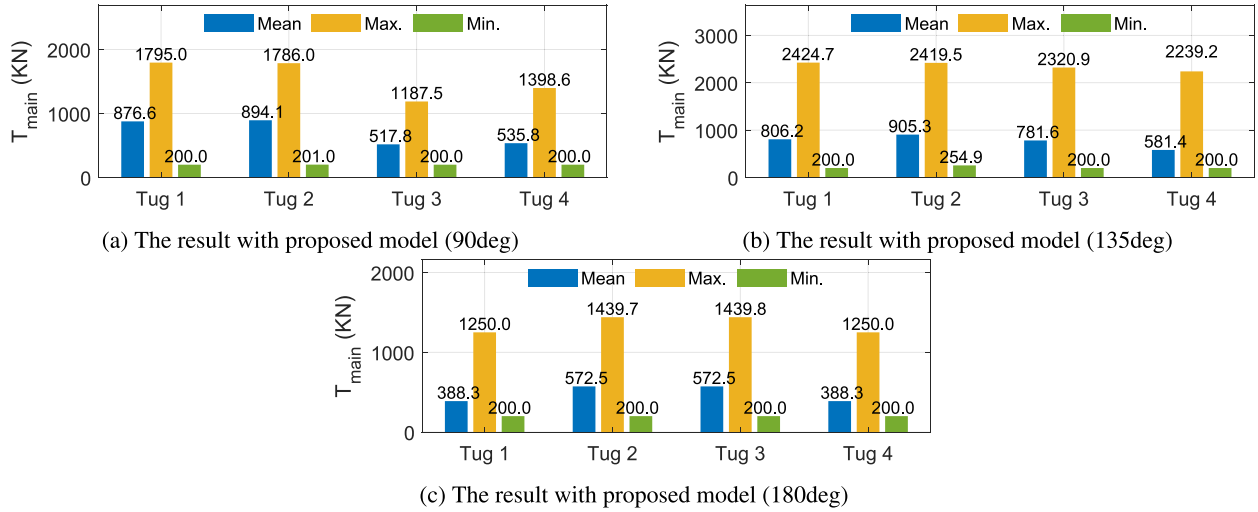


Fig. 16. The statistical results of $T_{main}^{t,j}$ with a real observer.

Table 14

The mean values of the τ_{main}^b with a real observer.

Direction	90 deg	135 deg	180 deg
Fx (kN)	-0.30	149.65	184.16
Fy (kN)	-621.04	-301.81	0.00
M (kN m)	1399.57	-3986.31	0.00

Table 15

The statistical results of thruster saturation with a real observer.

tugboat ID	90 deg		135 deg		180 deg	
	Count	Max.: $L_{\Delta T}$ (%)	Count	Max.: $L_{\Delta T}$ (%)	Count	Max.: $L_{\Delta T}$ (%)
1	1915	25.93	1878	16.39	2492	18.91
2	1577	23.66	2501	22.72	2555	24.64
3	1133	23.36	2189	10.15	2351	28.15
4	1586	24.70	1225	30.81	2266	18.91

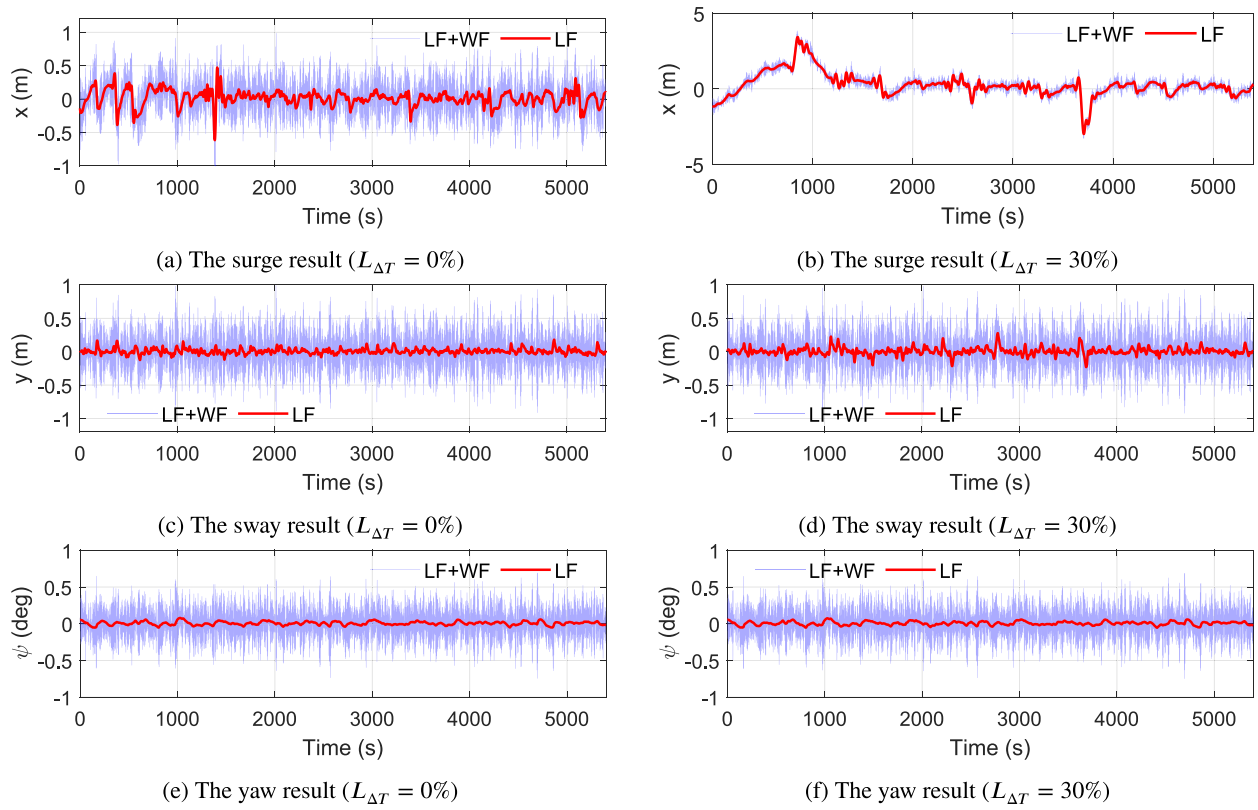


Fig. 17. The motion results of the barge with the conventional model and a real observer (135deg).

Table 16

The statistical results of the barge motion with the conventional model and a real observer ($L_{\Delta T} = 0\%$).

Direction		Mean	Max.	Min.	Std.	IAE (LF)
90deg	x (m)	0.000	0.096	-0.107	0.026	30.849
	y (m)	0.023	2.330	-1.992	0.591	584.012
	ψ (deg)	-0.008	0.183	-0.177	0.052	137.059
135 deg	x (m)	0.015	0.985	-1.311	0.261	464.201
	y (m)	0.007	0.991	-0.914	0.263	157.674
	ψ (deg)	0.006	0.692	-0.744	0.196	104.491
180 deg	x (m)	-0.004	0.794	-0.798	0.225	229.855
	y (m)	0.000	0.000	-0.001	0.000	0.157
	ψ (deg)	0.000	0.000	0.000	0.000	0.020

Table 17

The statistical results of the barge motion with the conventional model and a real observer ($L_{\Delta T} = 30\%$).

Direction		Mean	Max.	Min.	Std.	IAE (LF)
90 deg	x (m)	0.002	0.735	-0.360	0.082	212.098
	y (m)	0.023	2.850	-2.168	0.614	808.473
	ψ (deg)	-0.008	0.190	-0.174	0.052	137.059
135 deg	x (m)	0.269	3.838	-3.309	0.815	2961.451
	y (m)	0.006	0.980	-0.922	0.265	201.737
	ψ (deg)	0.006	0.689	-0.748	0.196	106.205
180 deg	x (m)	-0.003	0.865	-0.916	0.229	281.959
	y (m)	0.000	0.005	-0.017	0.001	0.918
	ψ (deg)	0.000	0.000	-0.001	0.000	0.073

4.2. The comparative analysis of the conventional model with the loss rate

In summary, numerical simulations with the conventional model must account for the loss rate. According to the statistical result, a fixed loss rate of 30% is applied, which means that the $\Delta T_{push}^{t,i}$ is constrained to $[-105, 105]$ kN/s. Comparative results for 135deg simulations with and without the loss rate are depicted in Fig. 17, with detailed statistics in Tables 16 and 17.

It is readily apparent that accounting for the loss rate degrades control effectiveness, indicating a potential need for further controller parameter tuning to optimize performance, though restoring original precision may not be guaranteed. With lower upper of the $\Delta T_{push}^{t,i}$, the probability of thrust

allocation errors rises. Employing a fixed loss rate to avoid main thrusters overload is inherently conservative, because saturation is neither constant nor always at peak loss rates. Nonetheless, in the absence of means to compute main thrust requirements using the conventional model, there seems to be no better solution, underscoring the advantages offered by the modified dynamics model.

5. Conclusion

This paper addresses the issue in pushing operations research where the uncontrollable directly pushing force is treated as the input to the dynamic model, which is always affected by unknown environmental disturbances. To resolve this, this paper have modified the existing LF mathematical model for the barge. On the premise that the relative positions of the tugboats remain unchanged, this paper introduces modification terms for the mass matrix, damping matrix, and environmental disturbances vector of the barge, based on the parameters corresponding to each tugboat.

Based on this, a modified dynamic model is proposed, while retaining the original LF dynamic model structure. It makes the motion response of the barge take into account the hydrodynamic of the barge and the tugboats as well as the influence of the environmental disturbances on them. At the same time, the controllable thrust generated by the main thrusters of the tugboats becomes the input of the dynamic model. The proposed model still does not require consideration of a collision model between the tugboats and barge, thereby avoiding an increase in model complexity and a decrease in computational efficiency.

Numerical simulations were conducted with different environmental disturbances direction to examine the dynamic positioning of barges pushed by tugboats. Using an ideal observer, the positioning accuracy achieved by the conventional model and the proposed model was comparable. Meanwhile, numerical simulations using both the unmodified and proposed models yield very similar results for the resultant force and moment generated by the pushing force, indicating that the pushing force is entirely devoted to counteracting environmental disturbances acting on the barge. In contrast, the thrust generated by the main thrusters of the tugboat is bigger, as it not only assists in positioning the barge but must also compensate for the environmental disturbances acting on the tugboat itself.

In addition, this study highlights the occurrence of thruster saturation during numerical simulations, wherein certain thrusters reached their operational limits at specific moments, yet the required changes of pushing force had not been met. Furthermore, the ideal observer was replaced with a real observer, and WF motions were introduced into the tug–barge system. The simulation results indicate that thruster saturation occurred more frequently, with a bigger loss rate observed in the changes of pushing force.

In summary, while the effectiveness of barge motion results derived from the conventional model is still acknowledged in this paper, it is not advisable to endow pushing force with the same performance parameters as the main thruster of the tugboats. Instead, a suitable loss rate should be applied to ensure that the expected push force remains achievable with environmental disturbances. Of course, this approach may compromise control system performance. Therefore, it is recommended that future pushing operation research incorporate the proposed model presented in this paper.

This paper has the following limitations : whether the pushing force generated by the tugboat is strictly along the direction of its captain needs further verification; the increased fluid interaction force of the tugboat due to its proximity to the barge is ignored; the change of the pushing angle of the tugboat is not discussed. Therefore, in future studies, the interaction force between the tugboat and the barge will be calculated by the elastic deformation of the rubber mat as well as the fluid interaction force, and the friction between the tugboat and the barge will also be considered. The results obtained will be compared with this paper.

CRediT authorship contribution statement

Ze Li: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yiting Wang:** Writing – review & editing, Visualization, Supervision, Software, Conceptualization. **Lei Wang:** Project administration, Methodology, Investigation, Conceptualization. **Ming Ding:** Writing – review & editing, Software, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

See [Table 18](#).

Table 18
The nomenclature table.

Parameter	Definition	Parameter	Definition
η_{target}	the target position of the DP vessel	$\mathbf{v}_{\text{target}}$	the target velocity of the DP vessel
η_{target}^b	the target position of the barge	$\mathbf{v}_{\text{target}}^b$	the target velocity of the barge
η_{ab}	the estimated LF motion of the DP vessel	$\mathbf{v}_{\text{target}}^b$	the estimated LF velocity of the DP vessel
η_{ab}^b	the estimated LF motion of the barge	$\mathbf{v}_{\text{target}}^b$	the estimated LF velocity of the barge
η_{sum}	the sum motion of the DP vessel	η_{WF}	the WF motion of the DP vessel
η_{sum}^b	the sum motion of the barge	η_{WF}^b	the WF motion of the barge
$T_i, T_a^{t,i}$	the allocated thrust	$\alpha_i, \alpha_a^{t,i}$	the allocated azimuth angle
$\mathbf{M}^b$	the generalized inertial matrix of the barge	$\mathbf{M}^{t,i}$	the generalized inertial matrix of the i th tugboat
$\mathbf{C}^{b, RB}$	the Coriolis matrix of the barge	$\mathbf{C}^{t,i, RB}$	the Coriolis matrix of the i th tugboat
$\mathbf{D}_L^b$	the linear damping matrix of the barge	$\mathbf{D}_L^{t,i}$	the linear damping matrix of the i th tugboat
$\mathbf{r}^b_{\text{env}}$	the environmental disturbances of the barge	$\mathbf{r}^{t,i}_{\text{env}}$	the environmental disturbances of the i th tugboat
$\mathbf{r}^b_{\text{push}}$	the resultant force and moment of the pushing force on the barge	$\mathbf{r}^{t,i}_{\text{thruster}}$	the resultant force and moment generated by the thrusters on the i th tugboat
$T^{t,i}_{\text{push}}$	the pushing force generated by the i th tugboat	$T^{t,i}_{\text{main}}$	the thrust generated by the main thrusters of the i th tugboat
$\mathbf{v}^b$	the velocity of the barge	$\mathbf{v}^{t,i}$	the velocity of the i th tugboat
$\mathbf{r}^b_{\text{wave2}}$	the second-order wave load on the barge	$\mathbf{r}^{t,i}_{\text{wave2}}$	the second-order wave load on the i th tugboat
$\mathbf{r}^b_{\text{wind}}$	the wind load on the barge	$\mathbf{r}^{t,i}_{\text{wind}}$	the wind load on the i th tugboat
$\mathbf{r}^b_{\text{current}}$	the current load on the barge	$\mathbf{r}^{t,i}_{\text{current}}$	the current load on the i th tugboat
$\mathbf{B}^b_{\text{push}}$	the direction matrix of the pushing force	$\mathbf{B}^{t,i}_{\text{main}}$	the direction matrix of the thrust generated by the main thrusters
$\alpha^{t,i}$	the direction of the pushing force	$L^{t,i}_G$	the distance from the center of gravity of the i th tugboat to its bow
$(L^{t,i}_x, L^{t,i}_y)$	the point of application for $T^{t,i}_{\text{push}}$	$(L^{t,i}_x, L^{t,i}_y)$	the point of the center of gravity of the i th tugboat
$\mathbf{r}^{t,i}_{\text{keep}}$	the control force for the sway and yaw motion of the i th tugboat	$\mathbf{R}$	the rotation matrix
η^b	the LF motion of the barge	$\eta^{t,i}$	the LF motion of the i th tugboat
$\eta^{t,i}_{\text{body}}$	the position of the i th tugboat relative to the barge	$x^{t,i}_G$	the distance from the center of gravity of the i th tugboat to its center
X^b_u, Y^b_v, Y^b_r	the added mass coefficients of the barge	X^b_u, Y^b_v, Y^b_r	the linear damping coefficients of the barge
N^b_u, N^b_v, N^b_r		N^b_u, N^b_v, N^b_r	
$X^{t,i}_u, Y^{t,i}_v, Y^{t,i}_r$	the added mass coefficients of the i th tugboat	$X^{t,i}_u, Y^{t,i}_v, Y^{t,i}_r$	the linear damping coefficients of the i th tugboat
$N^{t,i}_u, N^{t,i}_v, N^{t,i}_r$		$N^{t,i}_u, N^{t,i}_v, N^{t,i}_r$	
$\hat{\mathbf{y}}$	the estimated error	$\hat{\mathbf{y}}$	the estimated LF+WF motion
$\mathbf{K}_1, \mathbf{K}_2, \mathbf{K}_3$	the gain matrix of the NPO	$\hat{\eta}$	the estimated LF motion
$\mathbf{K}_4, \mathbf{A}_W, \mathbf{C}_W$		$\hat{\xi}$	the estimated WF motion
$\hat{\mathbf{b}}$	the estimated disturbances	$\mathbf{e}$	positioning error
s_{sf}	sliding surface	τ_{eq}	equivalent control force
$\mathbf{A}$	the gain matrix of the sliding surface	$\mathbf{K}_i$	the gain matrix for the integral term
τ_{sw}	switching control force	ϕ	boundary-layer thickness
$\mathbf{K}_{sw}$	switching matrix	$\mathbf{Q}$	weight matrix for slack variable
$\mathbf{W}$	weight matrix for thrust	$\Delta T_{\min}, \Delta T_{\max}$	the lower and upper bounds for thrust change rate
s	the slack variable	$L_{\Delta T}$	the loss rate of the thrust change rate
$T_{\min}, T_{\max}$	the lower and upper bounds for thrust	T_p	spectrum peak period
H_s	significant wave height	V_w	velocity of wind
V_c	velocity of current	Min.	minimum
Max.	maximum	IAE	the integral of absolute error
Std.	standard deviation	$C_{cx}, C_{cy}, C_{c\psi}$	the current load coefficients
$C_{wx}, C_{wy}, C_{w\psi}$	the wind load coefficients	U_{wr}/U_{cr}	the relative wind/current velocity
γ_{wr}/γ_{cr}	the relative wind/current direction	ω_i	the wave frequency
A_i	the wave amplitude	ζ_i	the wave elevation
ϕ_i	the initial wave phase	ϕ_{Pij}, ϕ_{Qij}	the phases of QTFs
P_{ij}, Q_{ij}	the QTFs		

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